GM UNIVERSITY

(Established under the Karnataka State Act No. 19 of 2023)
Post Box no-4, PB Road, Davangere-577006

Faculty of Engineering and Technology

Department of Computer Science and Engineering

Subject: Artificial Intelligence and Machine Learning

Subject Code: UE24CS2401

Question Bank

Module 1

Syllabus:

Introduction to Artificial intelligence:

What is AI: How Does AI Work, Advantages and Disadvantages of Artificial Intelligence

History of AI

Types of AI: Weak AI or Narrow AI, General AI, Strong AI/ Super AI, Reactive Machines, Limited Memory , Theory of Mind, Self-Awareness

Components of Intelligence: Agent and Environment,Types of agents

Informed Search Algorithms: Pure Heuristic, Best first, A*, AO* Algorithms.

CO2 Level Questions

1. Explain the concept of Artificial Intelligence with suitable examples.
2. Describe how Artificial Intelligence works in real-world applications.
3. Summarize the advantages and Disadvantages of Artificial Intelligence in modern society.
4. Summarize the major milestones in the evolution(history) of Artificial Intelligence.
5. Differentiate between Narrow AI and General AI.

CO3 Level Questions

1. Apply the Best First Search algorithm to solve a given graph problem and trace the steps.
2. Solve a pathfinding problem using the A* algorithm and explain how f(n) is calculated.
3. Apply the AO* algorithm to solve an AND-OR graph and identify the optimal solution path

CO4 Level Questions

1. Analyze the evolution of AI from its early stages to modern deep learning systems. What key technological changes influenced its growth?
2. Analyze how Weak AI differs from strong AI systems and what challenges exist in implementing it.
3. Analyze the different types of Agents in AI and its environment. Identify percepts, actions and goals with an example.

Module 3:

Module 3:

Introduction to Machine Learning : Need for Machine Learning, Machine Learning Explained

Machine Learning in Relation to other Fields: Machine Learning and Artificial Intelligence, Machine Learning, Data Science, Data Mining, and Data Analytics , Machine Learning and Statistics

Types of Machine Learning: Supervised Learning: Classification and regression, Unsupervised Learning, Semi-supervised Learning, Reinforcement learning, MACHINE LEARNING PROCESS, challenges, Applications

Understanding Data: what is Data (TYPES OF DATA), BIG DATA ANALYTICS AND TYPES OF ANALYTICS , BIG DATA ANALYSIS FRAMEWORK:DATA COLLECTION , DATA PREPROCESSING, Univariate Data Analysis and Visualization(Data Visualization, Central Tendency, Dispersion). Bivariate Data (Bivariate Statistics), Feature Engineering and Dimensionality Reduction Techniques (Step wise Forward and Backward).

CO2 Level Questions

1. Explain the need for Machine Learning in solving real-world problems.
2. Explain the relationship between Machine Learning and Artificial Intelligence.
3. Explain Supervised Learning with a suitable example
4. Describe the steps involved in the Machine Learning process.
5. Explain the differences between structured, semi-structured, and unstructured data with examples.
6. Explain the difference between descriptive, predictive, and prescriptive analytics.
7. Differentiate between univariate and bivariate data

CO3 Level questions

1. Apply the concept of data visualization to show how sales trends can be represented using Bar chart, Pie chart, Histogram, Dot Plots.
2. Apply the concept of dispersion for the patients list {12, 14, 19, 22, 24, 28, 31, and 34} and find IQR & Five point summary.

CO4 Level Questions

1. Analyze the role of Normalization procedure: Min-Max and Z-score in data preprocessing for given set V= {88, 90, 92, 94} for range 0-1 and V= {10, 20, 30} respectively.

2. Calculate Covariance and Correlation for the data X={1,2,3,4,5} and Y={1,8,27,64,125} using concept of Bivariate statistics.

Module 4:

Basic Learning Theory: Introduction to learning system, Design of Learning System

Introduction to Concept of Learning: Representation of Hypothesis, HypothesisSpace, Hypothsis space search by Find-S Algorithm, Limitations of Find-S Algorithm

Similarity-based Learning: introduction, Difference between Instance and Model Based learning, K Nearest Neighbor Algorithm, Weighted K-Nearest-Neighbor Algorithm, Numeric problems.

Regression Analysis: Introduction to Regression, Introduction to Linearity, Types of Regression methods, Limitations of Regression methods, Introduction to Simple Linear Regression and Numberic Problems

Decision Tree Learning: Introduction to Decision Tree Learning Model, Decision Tree Induction Algorithms ID3 Tree Construction

CO2 Level Questions

1. Explain the design of a learning system with suitable example.
2. Explain the Find's algorithms with its algorithm steps
3. Describe the concept of hypothesis space in machine learning.
4. Differentiate between instance-based learning and model-based learning.
5. Explain the steps involved in K-Nearest Neighbor algorithm and Liminations of KNN.
6. Describe Weighted KNN algorithm steps and its advantages over KNN.
7. Explain the concept of linearity in regression analysis.
8. Explain the structure of a decision tree learning model.

CO3 Level Questions

1. Apply Find-S algorithm to determine the most specific hypothesis for the given training dataset and list the limitations of it.

   Table 1

CGPA	Interactiveness	Practical Knowledge	Communication Skills	Logical Thinking	Interest	Job Offer
≥9	Yes	Excellent	Good	Fast	Yes	Yes
≥9	Yes	Good	Good	Fast	Yes	Yes
≥8	No	Good	Good	Fast	No	No
≥9	Yes	Good	Good	Slow	No	Yes

Table 2:

Example	Marks	Sports	Interview Performance	Extra Activities	Admission
1	High	Yes	Good	Yes	Yes
2	High	No	Good	Yes	Yes
3	Medium	Yes	Good	Yes	Yes
4	High	Yes	Average	No	No

2. Apply KNN algorithm for K=3 on the given dataset

3. Solve a classification problem using Weighted KNN algorithm where k=3

S.No	CGPA	Assessment	Project Submitted	Result
1	9.2	85	8	Pass
2	8.0	80	7	Pass
3	8.5	81	8	Pass
4	6.0	45	5	Fail
5	6.5	50	4	Fail
6	8.2	72	7	Pass
7	5.8	38	5	Fail
8	8.9	91	9	Pass

Training dataset and Test instance=(7.6,60,8) and (7.5, 70, 6) is same for both KNN and Weighted KNN (Qno 3 and Qno 4)

4. Construct a simple linear regression equation for the given dataset and predict the output value for 7th and 9th month sale.

Week (x_i)	Sales in Thousands (y_i)
1	1.2
2	1.8
3	2.6
4	3.2
5	3.8

CO4 level Questions

1. Compare and analyze KNN and Weighted KNN algorithms based on accuracy and performance.
2. Analyze different types of regression methods and their limitations in machine learning applications.
3. Analyze the effect of outliers on simple linear regression models.
4. Analyze the training data set and Construct a decision tree using ID3 algorithm

Table 1:

Sl No	Outlook	Play
1	Sunny	NO
2	Sunny	NO
3	Overcast	Yes
4	Rain	Yes
5	Rain	Yes
6	Rain	NO
7	Overcast	Yes
8	Sunny	NO

Table 2:

S.No	CGPA	Job Offer
1	≥9	Yes
2	≥8	Yes
3	≥9	No
4	<8	No
5	≥8	Yes
6	≥9	Yes
7	<8	No
8	≥9	Yes
9	≥8	Yes
10	≥8	Yes

Module 5:

Bayesian Learning: Introduction to Probability-based Learning, Fundamentals of Bayes Theorem and Classification Using Bayes Model: Naïve Bayes Algorithm with numeric example

Clustering Algorithms: Introduction to Clustering Approaches, Applications and challenges, K-means Algorithm with numeric examples

Artificial Neural Networks: Introduction, Biological Neurons, Artificial Neurons, ANN Structure, Activation Function, Perceptron and Learning Theory, Perceptron Algorithm, Types of Artificial Neural Networks

Performance Evalution matix : Classification, confusion matrix, accuracy, precision ,F1 score, Regression, Mean absolute error, Root Mean Square error

CO3 Level Questions

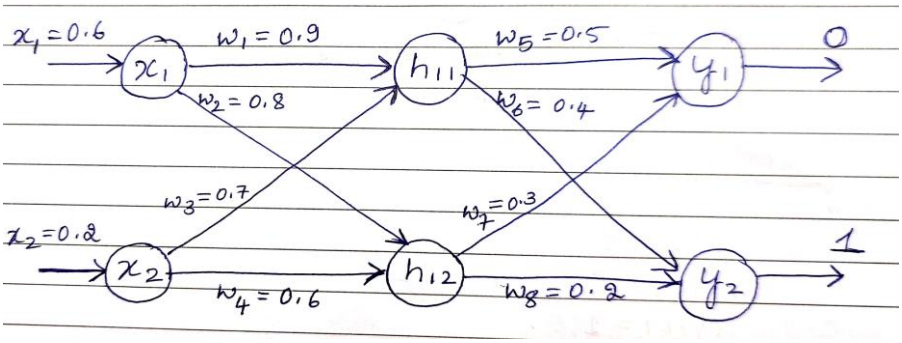
1. Apply the Naïve Bayes classification procedure and compute the posterior probability for a given dataset. Predict weather student gets the job offer or not.

S.No	CGPA	Interactiveness	Job Offer
1	≥9	Yes	Yes
2	≥8	No	Yes
3	≥9	No	No
4	<8	No	No
5	≥8	Yes	Yes
6	≥9	Yes	Yes
7	<8	Yes	No
8	≥9	No	Yes
9	≥8	Yes	Yes
10	≥8	Yes	Yes

2. Apply two iteration of the K-Means algorithm for a given dataset and initial centroids object 2 and object 5.

Object	X-coordinate	Y-coordinate
1	2	4
2	4	6
3	6	8
4	10	4
5	12	4

3. Calculate the output of an artificial neuron given input values, weights, bias=0.3, and activation function and target value.(Activation function : Sigmoid or Relu).



4. Calculate confusion matrix, accuracy, precision, recall, and F1-score for the given data set.

   A machine learning model is used to predict whether students will receive a job offer (Yes) or not (No). The following results were obtained for 100 students:

Actual / Predicted	Job Offer (Yes)	No Job Offer (No)
Job Offer (Yes)	45	5
No Job Offer (No)	10	40

CO4 Level Questions

1. Analyze the advantages and limitations of probability-based learning approaches.
2. Differentiate between Classification and Clustering.
3. Analyze the differences between biological neurons and artificial neurons.
4. Compare various activation functions used in neural networks.
5. Analyze the different types of neural networks with diagram.
6. Examine the structure and components of an Artificial Neural Network.
7. Analyze the impact of activation functions on ANN performance.

CO5 Level Questions

1. Differentiate between supervised learning and clustering approaches.
2. Assess the applications and challenges encountered while clustering high-dimensional data.

3. Evaluate the suitability of MAE and RMSE for house price prediction models.

   A machine learning model is used to predict house prices (in ₹ Lakhs). The actual and predicted prices are given below:

House	Actual Price (₹ Lakhs)	Predicted Price (₹ Lakhs)
H1	50	52
H2	65	60
H3	80	85
H4	100	95
H5	120	130

Module 2

Knowledge Representation: Introduction, Knowledge representation, Syntax, Semantics and logic inference,Cycle of Knowledge representation in AI , Knowledge based agent, The Architecture of Knowledge-Based Agent, Operations Performed By KBA, Types of knowledge, Techniques of knowledge representation in AI

Reasoning and Learning in AI: Reasoning, Features of a Reasoning System , Methods of Reasoning , Probabilistic Reasoning in AI.

CO4 Level Questions

1. Analyze the role of syntax, semantics, and inference in knowledge representation.
2. Compare declarative and procedural knowledge with suitable examples.
3. Analyze the architecture of a Knowledge-Based Agent and explain the role of each component.
4. Analyze the various types of knowledge used in AI systems.
5. Examine the cycle of knowledge representation and identify its importance in intelligent systems.
6. Analyze the operations performed by a Knowledge-Based Agent during problem-solving.
7. Analyze the various methods of Reasoning in AI.

CO5 level Questions

1. Critically evaluate the architecture of a Knowledge-Based Agent in solving AI problems.
2. Assess the importance of syntax, semantics, and inference in knowledge representation.
3. Critically examine different knowledge representation techniques and justify their applications.
4. Critically analyze the impact of uncertainty on AI reasoning systems.

5. Assess the suitability of various reasoning approaches for real-world problem-solving scenarios.
6. Evaluate the effectiveness of deductive, inductive, and abductive reasoning in AI applications.